Capstone Simulation — Final Run Sheet

Autonomous UAV target tracking · PX4 + Gazebo Harmonic + ROS 2 Jazzy · locked configuration, 9 Oct 2026

Before you start

Check	Command / action	Expect
GPU driver	`nvidia-smi`	RTX 4060 table, 60W cap
Power	Plug the laptop in	Cap reads 60W, not 15W or 30W
Nothing left over	`pgrep -a -f "gz sim"; pgrep -a -f px4`	No output
Latest scripts	`ls -la ~/Downloads/*.py`	No (1) copies left behind

Open the terminals in order and let each settle before the next. Terminal 1 must **not** have `rosenv` run in it.

Terminal 1 — PX4 + Gazebo

No rosenv here — sourcing ROS breaks gz sim.

```
cd ~/PX4-Autopilot
PX4_GZ_WORLD=city_test make px4_sitl gz_x500_gimbal
```

Then at the `pxh>` prompt:

```
param set COM_RC_IN_MODE 4
param set NAV_DLL_ACT 0
param set COM_DISARM_LAND 0
param set COM_DISARM_PRFLT 0
```

Terminal 2 — uXRCE-DDS agent

```
rosenv
MicroXRCEAgent udp4 -p 8888
```

Terminal 3 — Follower (flight control)

Click this window and hold SHIFT as soon as it prints 'Armed, offboard requested'. Keys only register while it has focus.

```
rosenv
python3 ~/Downloads/drone_follower.py --rate 10 --kp-yaw 0.3
```

Terminal 4 — ros_gz_bridge (camera + gimbal)

```
rosenv
ros2 run ros_gz_bridge parameter_bridge \
/world/city_test/model/x500_gimbal_0/link/camera_link/sensor/camera/image@sensor_msgs/msg/Image[gz.msgs
.Image \
/model/x500_gimbal_0/command/gimbal_pitch_demo@std_msgs/msg/Float64]gz.msgs.Double \
/model/x500_gimbal_0/command/gimbal_yaw_demo@std_msgs/msg/Float64]gz.msgs.Double \
--ros-args \
-r /world/city_test/model/x500_gimbal_0/link/camera_link/sensor/camera/image:=/x500/camera/image
```

Terminal 5 — YOLO detector

Source ROS first, then activate the venv. Order matters.

```
rosenv
source ~/ros_yolo_env/bin/activate
python3 ~/Downloads/sphere_detector.py
```

Terminal 6 — Camera view

```
rosenv
ros2 run rqt_image_view rqt_image_view /sphere_detector/image_annotated
```

Terminal 7 — Gimbal tracker

```
rosenv
python3 ~/Downloads/gimbal_tracker.py --kp 0.15 --deadzone 0.04 --hfov 2.0 --aspect 1.0
```

Terminal 8 — Position estimator + CSV logger

Add --slim for the eight-column file: t, gimbal_pitch_deg, est_x, est_y, est_vx, est_vy, true_x, true_y.

```
rosenv
python3 ~/Downloads/target_estimator.py --hfov 2.0 --aspect 1.0 --invert-gimbal-yaw --lag 0.8 --smooth
0.15 --vel-window 50 --max-speed 100
```

Running a flight

Step	Action
1	In terminal 3, hold SHIFT to climb to about 75 m
2	Fly toward the route (around x=739, y=150) with W/A/S/D
3	Wait until the gimbal locks on and terminal 8 prints est/true lines
4	Press F in terminal 3 to hand over to the follower
5	Press F again, or X, to take back manual control

Terminal 3 reads every keystroke while focused. Typing anything containing L lands the drone; F toggles follow; X stops. Click another window before typing there.

Locked settings and why

Setting	Value	Reason
`--hfov / --aspect`	2.0 / 1.0	Measured from the gimbal model: 115 degree FOV, 1024x1024. Wrong values corrupt every angle downstream.
`--invert-gimbal-yaw`	on	Camera is mounted rotated 180 degrees. Without it the estimate mirrors about the drone's heading.
`--lag`	0.8 s	Measured pipeline latency; projects the estimate forward.
`--smooth`	0.15	Position low-pass before differentiating.
`--vel-window`	50	Long enough to suppress velocity noise. Shorter reintroduced the rolling.
`--max-speed`	100	Only treats genuine target switches as jumps. At 25 it was clearing the velocity history on normal noise.
`--rate`	10 Hz	Matches the 10 Hz estimate, so commands aren't stepped.
`--kp-yaw`	0.3	Nose turns slowly toward the target: shares work with the gimbal through corners without chasing estimate noise. 0 starved the gimbal; 1.0 caused roll.
`--cmd-smooth`	0.1 (default)	Low-pass on the target velocity feedforward. The main cure for unnecessary roll.
Sphere speed	6.7 m/s	Pink route speed in server.config. Decoys remain at 10.

Shutting down

`shutdown` at `pxh>`, `Ctrl+C` in the others, then:

```
pkill -9 -f px4; pkill -9 -f "gz sim"; pkill -9 -f MicroXRCE; \
pkill -9 -f parameter_bridge; pkill -9 -f rqt; pkill -9 -f sphere_detector; \
pkill -9 -f gimbal_tracker; pkill -9 -f target_estimator; pkill -9 -f drone_follower

pgrep -a -f "gz sim"; pgrep -a -f px4
```

If something breaks

Symptom	Cause and fix
Script reports an unknown argument, or old behaviour after an update	The download saved as (1). `mv ~/Downloads/"x(1).py" ~/Downloads/x.py` then confirm with grep.
`No module named 'rclpy'`	ROS not sourced in that terminal. Run `rosenv`.
`nvidia-smi` fails	Driver not loaded. Reboot. If repeated, check `lsmod | grep nvidia` and `mokutil --list-enrolled`.
Gazebo GUI crashes, `GLXBadFBConfig`	GPU driver issue - see above.
Painfully slow	`nvidia-smi` shows low util or a 15/30W cap: plug in.
Drone lands and won't climb	Pressed L by accident, or disarmed. Restart terminal 3.
Arming denied	Re-run the four `param set` lines in terminal 1.
Everything frozen	Simulation paused. `Ctrl+Shift+P` in Gazebo.
Launch does nothing	Stale server. `pkill -9 -f "gz sim"`.
Spheres all pink, no route	Wrong plugin build. Check `grep -c ORANGE SphereNetworkSystem.cc` is above 0, rebuild.
Drone misses the final S-bend	Expected. 147 degree reversal after a 6 s leg; smoothness was chosen over responsiveness.

Data notes for the CSV

Item	Detail
Timebase	The t column is wall-clock. The simulation runs at about 75% of real time, so all speeds read ~25% below the configured value: the 6.7 m/s sphere logs at 5.0 m/s. Internally consistent, but not simulation time.
Irregular sampling	Rows are written only for accepted detections, so intervals vary (median 0.1 s, up to ~5 s). Any filter must use the actual time difference between rows, not assume 0.1 s.
Already filtered	est_x/est_y carry smoothing and 0.8 s lag compensation. For raw measurements run with `--smooth 1.0 --lag 0`.
Start-up	Discard the first second - the smoothing is still initialising.
Known fault	true_vx / true_vy / true_speed are unreliable: they read zero on ~17% of rows while the target is moving. Not in the --slim set. Derive true velocity from true_x/true_y instead.

Where things live

What	Path
Scripts	`~/Downloads/*.py`
Sphere routes + plugins	`~/PX4-Autopilot/src/modules/simulation/gz_bridge/server.config`
Sphere plugin source	`~/capstone/sim_city/sphere_system/SphereNetworkSystem.cc`
Trained model	`~/sphere_training/runs/sphere_n/weights/best.pt`
Flight logs	`~/target_log_<timestamp>.csv`
Backups	`~/server.config.sixgroups` · `SphereNetworkSystem.cc.before_waypoints`